REQUIREMENT SPECIFICATION & INTERVIEW REPORT

Hostel Attendance Management System

Project Name:	Hostel Attendance Management System
Assigned Agency / Team:	Spiders (Student Development Team)
Stakeholders Interviewed:	Director Sir, HOD (CSE), Mr. Mehra (Computer Science Dept), Chief Warden, Wardens
Document Type:	Functional Requirements & Director's Directive Alignment Report

1. Executive Summary

This report documents the detailed functional requirements, operational workflows, access permissions, and future scope for the proposed Hostel Attendance Management System. The project is assigned by Spiders and developed by the student development team under institutional guidance. This document synthesizes the base functional requirements with explicit directives, operational rules, and technical workflows agreed upon during the official interaction with the Director Sir and faculty representatives.

2. Key Discussion Directives & Strategic Decisions

During the meeting with the Director Sir, several operational and technical guidelines were clarified to ensure seamless execution, compliance, and multi-tier oversight across all hostels:

- **Absence & Defaulter Email Notification Hierarchy:** In the event of absence or failure to record attendance, notifications must be escalated sequentially to 6 key stakeholders: (1) Residential Warden, (2) Warden, (3) Chief Warden, (4) Director Sir, (5) Student's Parents, and (6) The Student.
- **Leave Application & Approval Workflow:** Students cannot directly apply for or approve leave in the system on their own. The student submits a formal application to the Residential Warden, who verifies and enters the leave into the system. Once logged by the Residential Warden, automated absence emails are suppressed for those approved dates.
- **Multi-Tier Dashboard Access & Cutoff Rules:** Role-based dashboards will be provided. Students can view their own attendance and leave records; Wardens see block/hostel-specific data; Chief Warden and Director Sir possess full administrative visibility over all hostels. Attendance freezes strictly at 7:00 PM IST daily, after which real-time defaulter lists (showing exactly who is missing as of 7:01 PM) become visible on executive dashboards.

- **Student Data Onboarding & Biometric Registration:** Students will not enter data. Caretakers/Officials will input verified profile details (Enrollment No, Name, Aadhaar, Room No, Parent Email, Contact). Students submit a registration form, which is then formally verified on the dashboard by the Warden.
- **Biometric Device Access & Server Integration:** Biometric devices store data locally without direct LAN connectivity. Mr. Mehra (Computer Science Dept) holds administrative access to fetch local data. The team is authorized to interface with central servers to push fetched logs into the central web portal DB.
- **Hostel Entry & System Testing Approvals:** To perform hardware verification, local data testing, and installation in hostels (including Girls' Hostel), student developers must be accompanied by Mr. Mehra / authorized technicians. Formal written authorization is issued via HOD CSE and Director Sir.
- **Formal Project Authorization Letter:** An official requisition letter issued through Spiders and forwarded by HOD CSE to Director Sir will formalize developer access to biometric hardware, student records, and institutional server credentials.

3. Functional Requirements Specification Table

ID	Feature	Detailed Requirement & Director's Modification
FR-01	Biometric Integration	Retrieve attendance logs locally/centralized via authorized scripts under Mr. Mehra's access.
FR-02	Student Management	Maintain enrollment ID, name, course, hostel room, email, parent email, and biometric ID. Data verified by Wardens.
FR-03	Parent/Guardian Management	Maintain parent contact details and email preferences for automated absence alerts.
FR-04	Hostel Management	Maintain hostel block, floor, room allocation, hostel allotment, and hostel empty (checkout) records.
FR-05	Attendance Processing	Apply 7:00 PM IST cutoff; mark present, late, leave, and trigger absent exception rules.

FR-06	Scheduled Execution	Run daily evaluation job at 7:00 PM IST, lock attendance entries, and log execution status.
FR-07	Defaulter Identification	Instantly flag non-marked/absent students at 7:01 PM for immediate follow-up.
FR-08	Email Notifications	Dispatch alerts sequentially to Residential Warden, Warden, Chief Warden, Director, Parents, and Student.
FR-09	Daily Reports	Generate hostel-wise, block-wise, and consolidated daily attendance
FR-10	Admin Dashboard	Provide customized role-based views for Student, Warden, Chief Warden, and Director Sir with real-time stats.
FR-11	Corrections & Leave	Residential Warden enters approved leave into system to override absence alerts; maintains audit history.
FR-12	Failure Handling	Detect network sync issues, stale biometric logs, missing parent emails, and notification delivery failures.
FR-13	Historical Reporting	Search, filter, and export historical attendance, leave records, and notification logs by date and hostel.
FR-14	Access Control	Enforce strict role-based authorization for administrative controls, data entry, and report viewing.

4. Future Scope & System Expansion Roadmap

Based on the strategic guidance provided during the interview, the project architecture will accommodate the following future expansion phases:

- **Complete Hostel Management Portal:** Integrate Hostel Allotment forms, Fee status tracking, and 'Hostel Empty' (vacating/checkout) formal applications directly into the web system.
- **Mess Management & Billing Integration:** Link student mess attendance, daily meal tracking, advance fee payments, and automated monthly rebate/billing generation.
- **Multi-Campus / Multi-College Deployment:** Expand system scalability to allow provision for other sister institutions and campus departments within the university system.
- **Per-Class Biometric Attendance (UGC Mandate Alignment):** Prepare system architecture to support classroom-level biometric hardware integration to satisfy UGC regulatory guidelines across academic blocks.